

22PH202 & ELECTROMAGNETISM AND MODERN PHYSICS

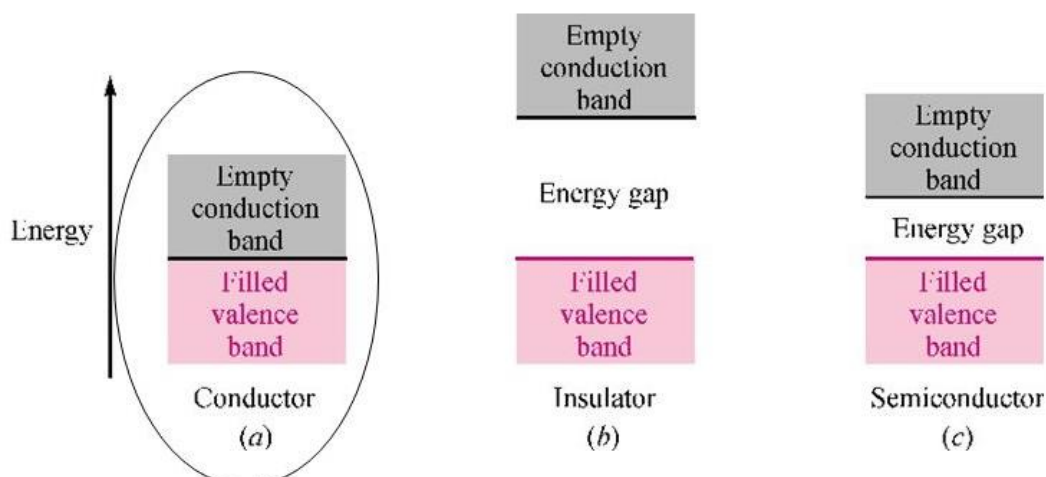
UNIT I – ELECTRICITY

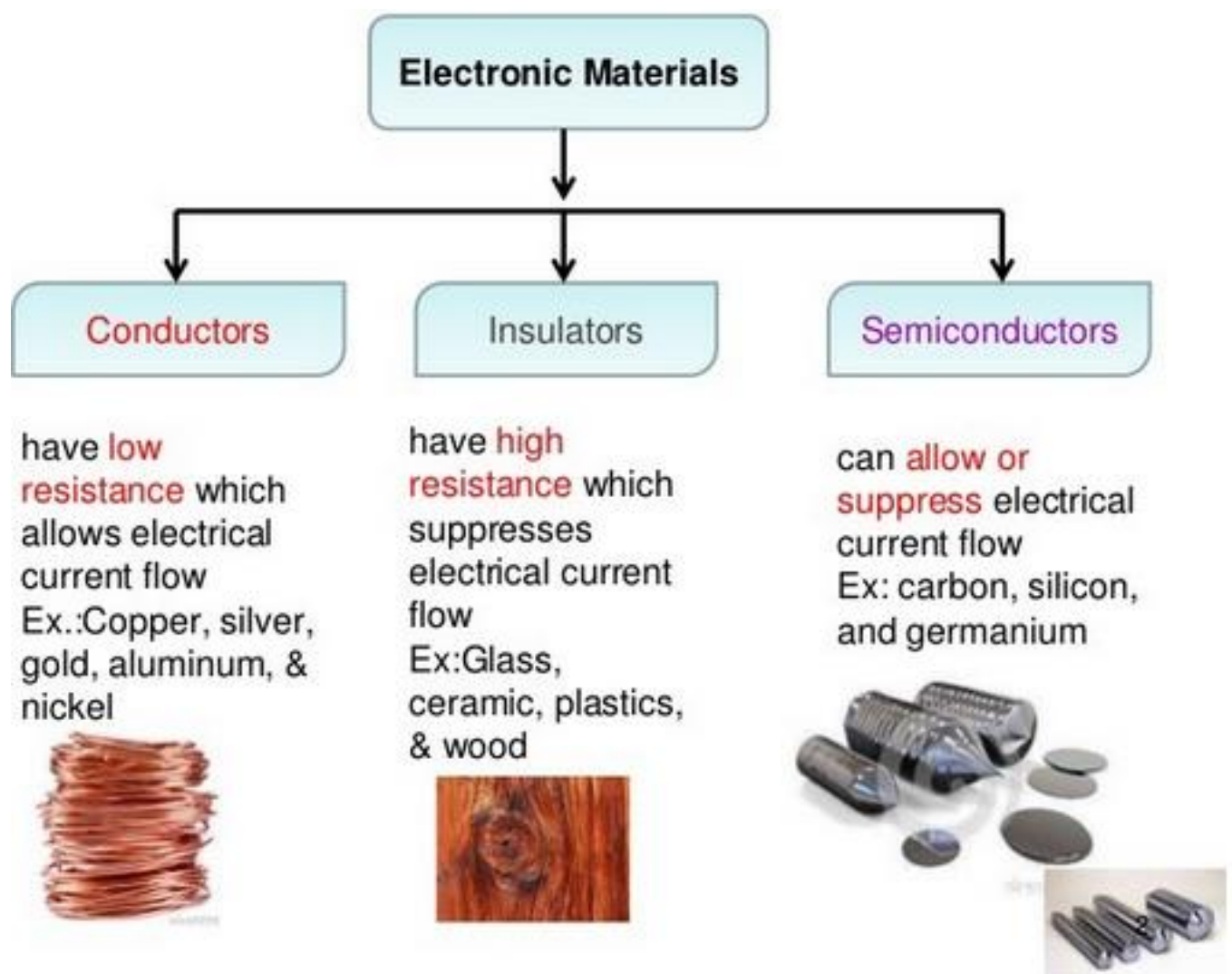
- Conductors
- Insulators



Introduction

- Conductors are those substances through which electricity can flow are called conductors.
- Insulators are those substances through which electricity cannot flow are called insulators.
- Some common conductors are copper, aluminum, gold, and silver.
- Some common insulators are glass, plastic, rubber, and wood.





Insulators and Conductors

Conductor:

Charge flows freely

Metals

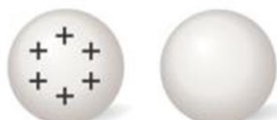
Insulator:

Almost no charge flows

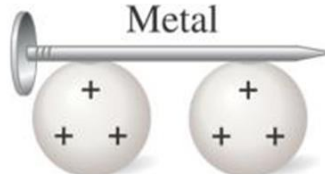
Most other materials

Some materials are semiconductors.

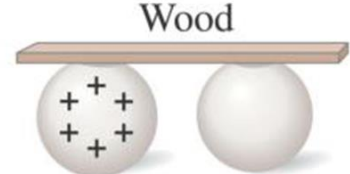
Charged Neutral

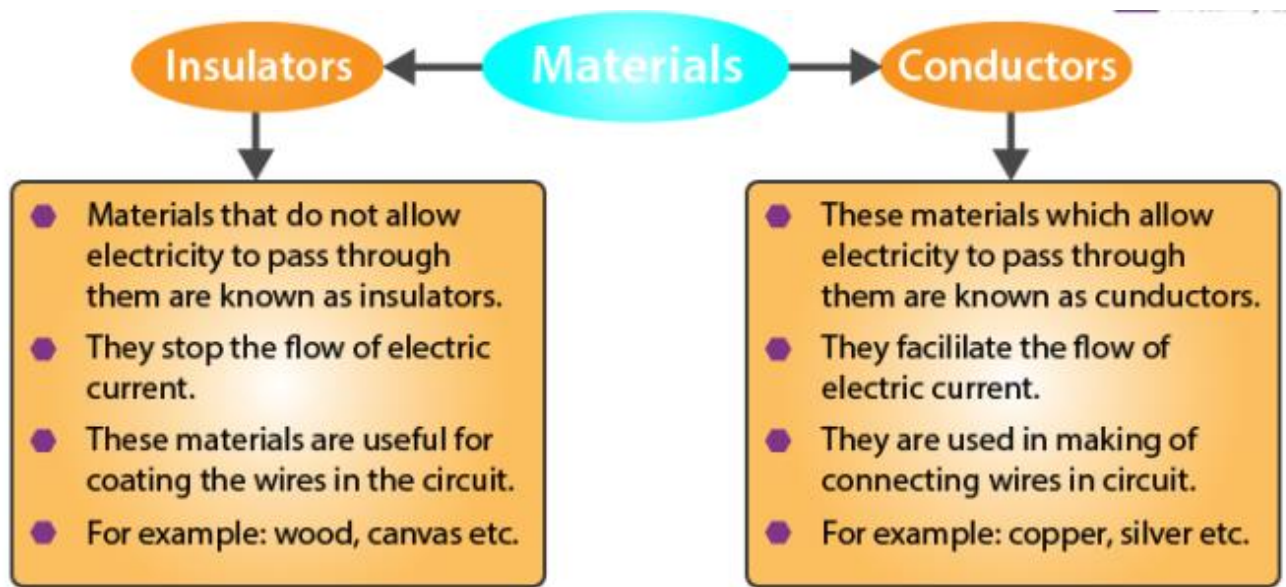


Metal

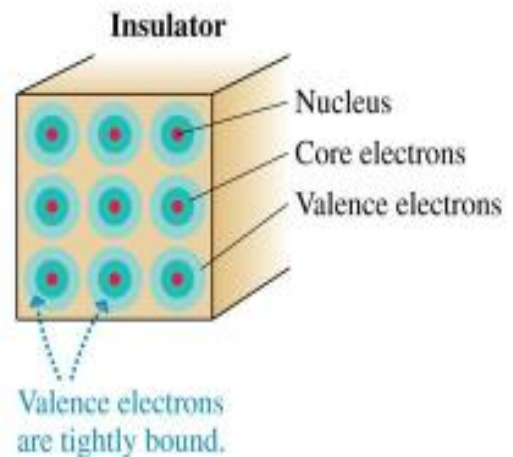


Wood

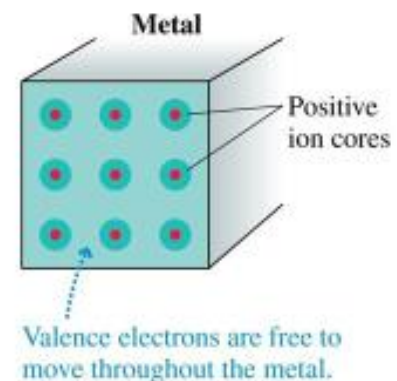




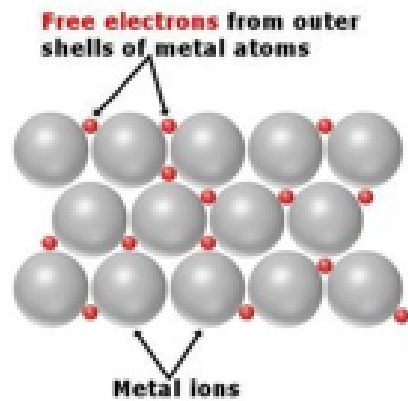
- The electrons in an insulator are tightly bound to the positive nuclei and are not free to move around.



- Charging an insulator may leave a patch of molecular ions on the surface, but the patches are immobile.
- In metals, the outer electrons (*valence electrons*) are weakly bound to the nuclei.
- They are detached from their parent nuclei and are free to wander through the entire solid, creating a **sea of electrons** permeating an array of positively charged **ion cores**.



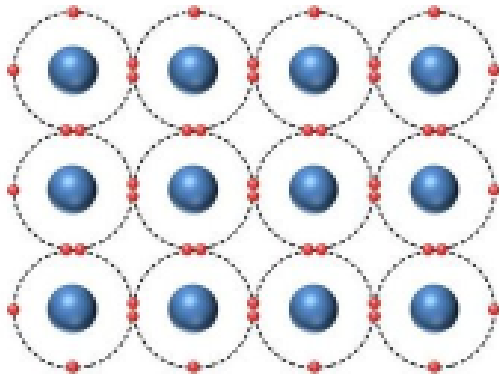
Conductors and Insulators



Conductors

Some of the electrons are not free electrons (not bound to atoms)

Free electrons can move relatively freely through the materials



Insulators

All the electrons are bound to atoms

Electrons cannot move freely through the material

Conductors and insulators

- **Conductors** - materials with little resistance to the flow of electric charge (e.g. plastic coated copper wires used in most electric circuits = excellent conductors).
- Little energy is needed for electric charge to flow through conductors.
- Most metals are very good conductors of electricity.

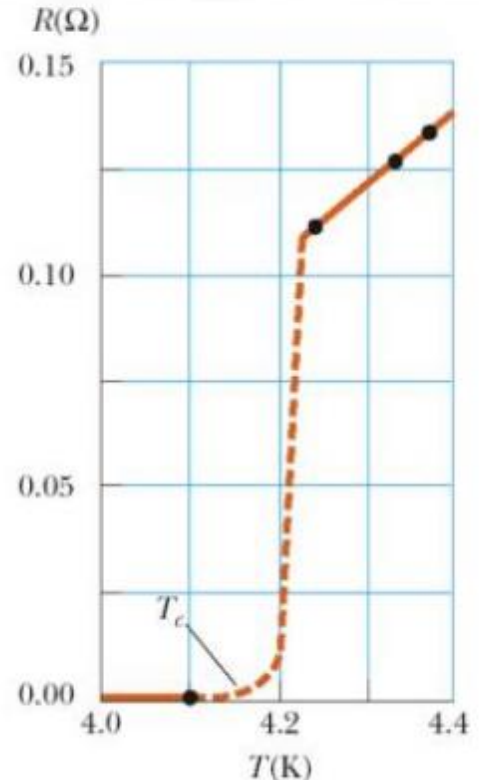


- Charge can be transferred from one object to another only when the objects *touch*.
- **Discharging** is removing a charge from an object, which you can do by touching it.

- **Conductors** are those materials through or along which charge easily moves.
- **Insulators** are materials in which charge is immobile.
- Glass and plastics are insulators; metal is a conductor.

Superconductors

- A class of materials and compounds whose resistances fall to virtually zero below a certain temperature, T_C
 - T_C is called the critical temperature
- The graph is the same as a normal metal above T_C , but suddenly drops to zero at T_C



Applications

- **Superconducting levitation:**
Maglev Trains (On Track with superconductivity)



Summary

- Electrical conductors: They allow electric current to flow easily.
- In general, the best electrical conductors are metals.
- The best conductor is silver, but it is too rare and expensive.
- The most commonly used electrical conductor is copper.
- Pure water does not conduct electricity.
- Superconductors: A perfect conductor with electrical resistance of zero at -234°C .
- Electrical insulators: Materials that do not transfer electricity.
- Protect us from getting shocked.
- Examples: wood, plastic, rubber, wool, silk, fur, oil, paper.
- Most good electrical conductors are also good conductors of heat.